XAVIER DIAS

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Professional Summary

Embedded Software Engineer with **2 years 5 months** of experience in developing **real-time, safety-critical aerospace software** following **DO-178C** processes. Experienced in **bare-metal and Embedded Linux development** on **ARM Cortex-A9, Cortex-M4, and Cortex-M7** platforms. Strong expertise in **Embedded C, C++, and Python**, low-level software development, bootloader implementation, memory management, peripheral driver development, and avionics communication protocols. Hands-on experience with **ARINC 429/739**, DMA, SDRAM, watchdog timers, interrupt handling, inter-core communication, and software verification using **LDRA**. Skilled in developing robust embedded solutions for aerospace display, navigation, and flight data logging systems.

Technical Skills

- **Programming Languages:** Embedded C, C++, Python
- **Embedded Platforms:** STM32 (Cortex-M4/M7), TMS570LC4357, i.MX6Q (Cortex-A9), LPC3250
- **Operating Systems:** Bare-metal, Embedded Linux
- **Protocols:** ARINC 429, ARINC 717, ARINC 739, SPI, I2C/I3C, UART, USB, TCP/UDP, IPCC
- **Standards:** DO-178C, MISRA C/C++
- **Tools:** STM32CubeIDE, LDRA, Astronics Ballard CoPilot, J-Flash, PyCharm, OpenGL, PyQt5
- **Version Control / Traceability:** Git, GitHub, SVN, TortoiseSVN
- **Debugging Tools:** Oscilloscope, JTAG Debugging, Lab Validation Setups
- **Cloud / Certifications:** AWS Certified Cloud Practitioner

Work Experience

Engineer Software — Alten India, Bengaluru **Jan 2024 – Present**

- Developed real-time aerospace embedded software for avionics display, navigation, and flight data recording systems following **DO-178C** software lifecycle processes.
- Implemented low-level embedded modules involving memory initialization, peripheral communication, interrupt-driven interfaces, and real-time data acquisition.
- Contributed to software architecture, design documentation, verification activities, and code quality analysis for safety-critical aerospace applications.
- Performed software validation, debugging, and hardware integration using oscilloscopes, STM32CubeIDE, JTAG debugging, and laboratory test environments.
- Worked on **LDRA-based code analysis, code coverage measurement, and software verification review activities**.
- Developed Python automation tools for pseudo-code generation, traceability mapping, and test workflow support.

Projects

1. Multi-Control Display Unit (MCDU) – Avionics Display Interface

Technologies: Embedded C, C++, Python, PyQt5, DLL, REST API, Ballard SDK, TCP/UDP Sockets, ARINC 429/739, TMS570LC4357

- Developed a real-time **Multi-Control Display Unit (MCDU)** for avionics subsystems in a safety-critical aerospace environment.
- Implemented a **bootloader** for the **TMS570LC4357** platform to enable reliable firmware upload and update mechanisms.
- Developed low-level memory handling modules involving **Flash, RAM initialization, memory mapping, and efficient memory utilization** in a bare-metal environment.

- Implemented **ICD-based communication** between analog hardware and processor using **MibSPI with DMA**, ensuring reliable and deterministic data transfer.
- Integrated Python-based GUI applications using **PyQt5** with C/C++ DLL interfaces for high-speed command and telemetry exchange.
- Integrated **Astronics Ballard CoPilot APIs** for real-time **ARINC 429** communication and avionics data exchange.
- Developed Python automation scripts for **ARINC label exchange testing** using Ballard tools.
- Authored aerospace software documentation including software design and development artifacts aligned with DO-178C processes.
- Contributed to software architecture discussions and high-level component design.

2. Real-Time Navigation System

Technologies: Embedded C, Python, OpenGL, ARM Cortex-A9, i.MX6Q, Flash, SDRAM, Embedded Linux

- Developed embedded navigation software for an **ARM Cortex-A9 (i.MX6Q)** based aerospace platform.
- Worked on hardware-near software involving **SDRAM access, memory-mapped peripherals, framebuffer handling, and flash memory interfaces**.
- Performed cross-compilation, deployment, and target-level debugging using JTAG-based debugging workflows.
- Implemented system-level optimizations for reliable **real-time GPS data acquisition and processing**.
- Developed Python-based tooling to convert **GeoJSON and image assets into binary data formats** for navigation rendering pipelines.
- Developed graphical rendering components using **OpenGL** to improve navigation display performance and responsiveness.

3. Flight Data Recorder (FDR) – Avionics Logging System

Technologies: Embedded C, Embedded C++, STM32 Cortex-M4/M7, ARINC 429/717, SPI, USB, Flash, I2C, TCP/UDP

- Contributed to the development of a **Flight Data Recorder (Black Box)** for aerospace data acquisition and logging applications.
- Developed real-time data acquisition modules interfacing with **HOLT avionics ICs** for protocol communication.
- Implemented a **multi-channel audio acquisition system** on a dual-core STM32 platform using **IPCC inter-core communication and shared memory buffering**.
- Developed embedded modules for reliable high-throughput data transfer between acquisition and storage subsystems.
- Worked on embedded communication interfaces involving **SPI, USB, Flash memory, and TCP/UDP-based transfer mechanisms**.
- Supported software verification through **LDRA analysis, code coverage measurement, and validation activities**.
- Authored software design and verification documentation for aerospace software development.

Education

Bachelor of Engineering (Computer Science and Engineering) 2019 – 2023

Jain College of Engineering and Research CGPA: 8.42

Competitive Programming

- Solved 350+ algorithmic problems on LeetCode
- Maximum LeetCode Rating: 1800